DLS Buddy — Advanced Guide (Theory & Equations)

Version 0.9.0

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Contents

Equations are numbered (1), (2), … and rendered in LaTeX. Whenever a new analysis is added to the program, its equation(s) are added here.

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7. Physics and Calibration

All physical formulas live in `physics/constants.py`; the analysis code never re-implements them.

7.1 Scattering Vector

$$q = \frac{4\pi n}{\lambda} \sin\left(\frac{\theta}{2}\right) \tag{3}$$

where λ is the vacuum wavelength of the laser, n is the solvent refractive index at that wavelength and temperature, and θ is the scattering angle. Both λ and n are always user-supplied. q is computed in nm^{-1} for SLS and in m^{-1} for the Stokes–Einstein step. This is the standard definition for solution scattering [1].

7.2 Stokes–Einstein Relation

The hydrodynamic radius follows from the translational diffusion coefficient $D = \Gamma/q^2$ (Γ is the field-autocorrelation decay rate) via the **Stokes–Einstein relation** [1]:

$$R_h = \frac{k_B T}{6\pi\eta D} \tag{4}$$

where kB is the Boltzmann constant, T the absolute temperature, and η the solvent dynamic viscosity. T and η are always user-supplied. A single-angle measurement gives the **apparent** Rh; extrapolation to $q \to 0$ and $c \to 0$ is handled in the analysis module.

7.3 Refractive-Index Correction

When the sample solvent differs from the calibration standard, the excess Rayleigh ratio is corrected by

$$f = \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}}\right)^2 \tag{5}$$

applied as $\Delta\mathrm{R}\theta = \mathrm{R}\theta\text{measured}\cdot f$. When solvent = standard, $f = 1$ exactly. This correction is **mandatory** whenever the solvent is not the calibration liquid.

7.4 Optical Constant

The prefactor in the Zimm/Debye/Berry equations (vertically polarized geometry):

$$K = \frac{4\pi^2 n^2 \left(\mathrm{d}n/\mathrm{d}c\right)^2}{N_A\,\lambda^4} \tag{6}$$

with NA the Avogadro number. Units are $\mathrm{mol}\cdot\mathrm{mL}^2\cdot\mathrm{g}^{-2}\cdot\mathrm{cm}^{-1}$ (λ converted nm→cm internally), consistent with c in g/mL and $\mathrm{R}\theta$ in cm^{-1}.

Why dn/dc matters. $K \propto (\mathrm{d}n/\mathrm{d}c)^2$, so the refractive-index increment enters the molecular weight **squared**. For low-contrast systems (e.g. PVP in reline) a small dn/dc makes the scattering weak and amplifies its error. It is user-supplied and must never be guessed.

7.5 Toluene Rayleigh Ratio (geometry-aware)

All SLS calibration uses the temperature- and geometry-corrected toluene Rayleigh ratio — **never** the value stored in instrument software. The base quantity is the polarized ratio RVV, temperature-corrected and then converted to the measurement geometry:

$$R_{VV}(T) = R_{VV}(25°\mathrm{C})\,[\,1 + \alpha\,(T - 25)\,], \qquad \alpha = +0.0043\ °\mathrm{C}^{-1} \tag{7a}$$

$$R_{VU} = R_{VV}\,(1 + \rho_v), \qquad R_{VH} = R_{VV}\,\rho_v \tag{7b}$$

where ρv = IVH/IVV is the depolarization ratio, ρv(T) = 0.346 − 0.00135 (T − 25) unless overridden.

Wavelength	RVV at 25 °C (cm^{-1})	Source
532 nm	2.34 × 10^{-5}	[2]
660 nm	8.456 × 10^{-6}	[3], Table 2

Geometry must match the calibrant. The temperature coefficient α is **positive** — the Rayleigh ratio *rises* with temperature. The geometry (VV / VU / VH) must match how the calibrant was measured, or the absolute scale is wrong by ≈ (1 + ρv) ≈ 1.35. An instrument with a vertical laser and **no analyzer** is **VU** geometry. Example: RVU(532 nm, 25 °C) = 2.34 × 10^{-5} × (1 + 0.346) ≈ 3.15 × 10^{-5} cm^{-1}.

Other wavelengths. Add a properly sourced RVV to `_TOLUENE_RVV_25C` in `physics/constants.py`. Do **not** interpolate — the wavelength dependence is steep (≈ λ^{-4}.^{2}).

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9. Data-Quality and Cross-Analysis Theory

9.1 Intensity-trace diagnostics

Two quantities reported by the trace diagnostics (the procedure is in the Quickstart) rest on the shot-noise model of photon counting.

Outlier band. Points outside the shot-noise envelope are flagged:

$$\text{flag if } |I - \bar{I}| > k\sqrt{\bar{I}}, \qquad k = 1 \text{ (default)} \tag{8}$$

For a Poisson (shot-noise-limited) process the standard deviation equals $\sqrt{\text{mean}}$, so this is the expected $\pm k\sigma$ band; the flagged fraction is a compact quality metric. Note $\sqrt{\text{mean}}$ equals the true shot-noise σ only for raw photon counts; for count rates (cps) the envelope is a useful heuristic.

Fano factor. The count-rate histogram is fitted with a Gaussian and/or Poisson distribution; the key diagnostic is the Fano factor (variance/mean in count space):

$$F = \frac{\sigma^2}{\mu} \quad \begin{cases} \approx 1 & \text{shot-noise limited (ideal Poisson)} \\ > 1 & \text{excess fluctuations (slow modes, dust, drift)} \end{cases} \tag{9}$$

9.4 $\rho = R\mathrm{g}/R\mathrm{h}$

The shape parameter links the radius of gyration (Rg, from SLS) and the hydrodynamic radius (Rh, from DLS) of the same sample:

$$\rho = \frac{R_g}{R_h} \tag{10}$$

Both should be infinite-dilution values for ρ to be meaningful; the program checks that the two measurements share polymer, solvent, and temperature. The architecture reference values below are well-known guides, not sharp boundaries [1].

ρ	Architecture
≈ 0.775	Hard sphere (compact, near-spherical)
≈ 1.0	Hyperbranched polymer, microgel, or vesicle
≈ 1.5–1.8	Linear random coil (θ-solvent ≈ 1.5, good solvent ≈ 1.78)
> 2	Extended/rigid-rod, or aggregated/polydisperse

These are guides, not hard boundaries. **ρ is flagged apparent if *either* input is apparent** — a single-condition Rg or Rh is never silently combined with a thermodynamic one. For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

9.5 Scaling laws (Rg–Mw, A_2–Mw)

Across a homologous series (one polymer/solvent, varying molecular weight), size and the second virial coefficient follow power laws in Mw:

$$R_g = k_g \, M_w^{\nu} \tag{28}$$

$$A_2 = k_a \, M_w^{-a} \tag{29}$$

The program fits each as a straight line in log–log space; the **exponent** is the physically meaningful quantity and is independent of the (calibration-dependent) prefactor. Typical flexible-coil values: $\nu \approx 0.588$ in a good solvent, $1/2$ at the θ-point, $1/3$ for a collapsed globule or hard sphere; $a \approx$ 0.2–0.3 [4], [5]. The fit needs at least two samples with the relevant pair. Below each plot the tab prints a **plain-language reading** of the fitted exponent (e.g. "$\nu \approx 3/5 \rightarrow$ swollen coil, good solvent"), mirroring the ρ table's architecture hint — a guide, not a hard boundary.

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10. Dynamic Light Scattering (DLS) Analysis

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau){-}1$. All methods fit $g_2(\tau){-}1$ **directly** (β floating) and carry the Siegert factor of 2 explicitly, so a reported decay rate Γ is the physical g_1 rate; then $D = \Gamma/q^2$ and Rh follows from Stokes–Einstein, Eq. (4) (which needs the viscosity; decay rates and diffusion coefficients do not).

The two correlation-to-quantity conversions used below appear first as data forms when loading a generic file. For the `g2` form, the baseline is removed by

$$g_2(\tau) - 1 = \frac{G(\tau) - B}{B} \tag{1}$$

For the `g1` form, the **Siegert relation** [1] converts the field autocorrelation to the measured quantity:

$$g_2(\tau) - 1 = \beta\,|g_1(\tau)|^2 \tag{2}$$

where β is the instrument coherence factor (an optical property, typically 0.01–1.0 for standard goniometers), determined independently.

10.1 Parametric Methods

Cumulant expansion [6]**.** The standard first analysis. Taking the log of the baseline-subtracted signal and fitting a polynomial in τ:

$$\ln\!\big[g_2(\tau) - 1\big] = \ln\beta - 2\bar{\Gamma}\,\tau + \mu_2\,\tau^2 - \tfrac{1}{3}\,\mu_3\,\tau^3 + \cdots \tag{11}$$

The mean decay rate $\bar{\Gamma}$ gives $\bar{D} = \bar{\Gamma}/q^2$ and Rh; the second cumulant gives the **polydispersity index**

$$\mathrm{PDI} = \frac{\mu_2}{\bar{\Gamma}^2} \tag{12}$$

and the third (order-3 fit) the skewness. The cumulant result is reliable only up to PDI $\approx$ 0.3; the program flags whether this is satisfied. Above it, use a distribution method. Equation (11) is the **linear** (log-polynomial) method; it assumes the baseline is exactly zero and is fit by weighted linear least squares.

Nonlinear cumulant [7]**.** The program's **default** cumulant method fits $g_2{-}1$ *directly* by nonlinear least squares, expanding about the mean rate (Frisken's moments-about-the-mean form) with a **floating baseline** B:

$$g_2(\tau) - 1 = B + \beta\Big[e^{-\bar{\Gamma}\tau}\big(1 + \tfrac{1}{2}\mu_2\tau^2 - \tfrac{1}{6}\mu_3\tau^3 + \cdots\big)\Big]^2 \tag{47}$$

It yields the same $\bar{\Gamma}$, μ_2, PDI and Rh as the linear method but is more robust to baseline drift and noisy/low-count data, because B and the coherence β are fit rather than assumed and the whole windowed correlogram is used (no amplitude cutoff). Expanding about $\bar{\Gamma}$ (rather than about $\tau = 0$, as the log-polynomial does) keeps it stable at large τ. The baseline is constrained to small drift about the long-time tail so the fit cannot absorb the decay into B; if it fails to converge it falls back to the linear fit, flagged. The method is a **global setting** (**Settings → Cumulant method**, default *Nonlinear*) that applies to every cumulant-based step (per-measurement cumulant, Γ-vs-q^2, replicate averaging); changing it clears existing cumulant-based results so the workspace stays self-consistent. **Uncertainty:** like the linear method, the nonlinear fit reports **no ± from a single correlogram** (its lag channels are correlated — Schätzel 1990 — so the fit covariance under-reports); the honest DLS ± comes from replicate averaging (Eq. 37), per invariant 8.

Analysis window: delay bounds + leading-channel skip. Every fit (cumulant, exponentials, KWW, NNLS/CONTIN/lognormal) selects which correlogram points it uses through one shared window with two complementary restrictions, applied in order:

1. a **leading-channel skip** — drop the first N correlator channels *by index* (the shortest, finest-spaced lags). The first few channels of a multi-τ correlator carry detector **afterpulsing** and dead-time artefacts that are not diffusion: a spuriously high first channel mimics a fast decay and biases the fit toward a falsely small Rh. The skip is an integer set in **Settings → Skip initial lag channels** (default 0 = keep all);
2. a physical **delay-time window** [τmin, τmax] in time.

The two **compose as an intersection** — the effective first fitted point is the later of "channel N" and "first channel with $\tau \geq \tau$min". The skip is applied **first**, so a corrupted leading channel cannot distort the amplitude cutoff/intercept (cumulant) or the β-to-$\tau \to 0$ extrapolation (distributions). Use it sparingly with distribution methods: the short lags constrain the *fast / small* end of the size distribution, so an aggressive skip that a cumulant *mean* merely nudges can erase a genuine small-particle population entirely.

Single exponential — a monodisperse sample (equivalent to the first cumulant for clean data):

$$g_2(\tau) - 1 = \beta\, e^{-2\Gamma\tau} \tag{13}$$

Double exponential — two discrete modes through the full Siegert relation (appropriate when a polymer and a slow aggregate coexist); the fit reports whether it converged:

$$g_2(\tau) - 1 = \beta \left[\, f_1\, e^{-\Gamma_1\tau} + (1 - f_1)\, e^{-\Gamma_2\tau} \,\right]^2 \tag{14}$$

Kohlrausch–Williams–Watts (stretched exponential) — broadly distributed or glassy dynamics, stretch exponent $s \in (0,\, 1]$ (s = 1 recovers a single exponential):

$$g_2(\tau) - 1 = \beta\, \exp[-2\,(\tau/\tau_c)^{\,s}]\,, \qquad \langle\tau\rangle = \frac{\tau_c}{s}\,\Gamma\!\left(\frac{1}{s}\right) \tag{15}$$

Both the characteristic time τc and the mean time $\langle\tau\rangle$ (via the gamma function) are reported, each with its hydrodynamic radius.

10.2 Distribution Methods

These recover a full distribution of decay rates without assuming a functional form. The field autocorrelation is a weighted sum of exponentials,

$$g_2(\tau) - 1 = \beta \left[\int G(\Gamma)\, e^{-\Gamma\tau}\, \mathrm{d}\Gamma \right]^2, \tag{16}$$

discretized on a grid of hydrodynamic radii (default 1–1000 nm, log-spaced) as $\mathbf{A}\ \mathbf{x} = \mathbf{y}$. Both methods need the coherence factor β and a baseline (estimated automatically — β by extrapolating to zero delay, the baseline from the long-delay tail — and overridable; the values used are always reported).

- **NNLS** solves $\min\|\mathbf{Ax} - \mathbf{y}\|^2$ with $\mathbf{x} \geq 0$ and no smoothing. It resolves well-separated peaks but produces spiky, noise-sensitive distributions.
- **CONTIN** [8] adds Tikhonov regularization:

  $$\min_{\mathbf{x}\geq 0} \|\mathbf{Ax} - \mathbf{y}\|^2 + \alpha^2 \|\mathbf{Lx}\|^2 \tag{17}$$

  with $\mathbf{L}$ the second-difference operator. The regularization strength α is chosen by an L-curve criterion (the full sweep is returned and a fixed α can be supplied). CONTIN is generally better for broad distributions and real data.
- **Lognormal** is a *single-mode parametric* distribution: it assumes the intensity-weighted size distribution is lognormal in Rh and fits its median and log-width to the same $\mathbf{A}\ \mathbf{x} = \mathbf{y}$ model. On the log-spaced grid the discrete weight is $w_i \propto \exp[-(\ln R_{h,i} - \mu)^2/2\sigma^2]$ with $\mu = \ln(\text{median } R_h)$. It cannot resolve multiple modes, but it is robust to noise and always returns a smooth, single-peaked distribution — a good default for a known-unimodal sample.

  Caution. Fine structure — small secondary peaks especially — is sensitive to noise and the regularization choice and is not always reproducible. Treat the main peaks as reliable; cross-check width against the cumulant PDI and the parametric fits.

10.3 Combining Measurements

Γ versus q^2 (multiple angles). For pure translational diffusion the decay rate is proportional to q^2:

$$\bar{\Gamma} = D\, q^2 \tag{18}$$

A straight line through the origin gives D (and Rh). Three linearity diagnostics test whether the motion is genuinely diffusive: the fit R^2; the intercept of an unconstrained fit (should be ≈ 0 — a nonzero value indicates a non-diffusive contribution); and the trend of Dapp $= \Gamma/q^2$ with q^2 (should be flat — an upturn signals internal/segmental modes). A combined diffusive / non-diffusive flag summarizes these.

Concentration extrapolation. Extrapolating the apparent diffusion coefficient to zero concentration removes inter-particle interactions:

$$D_{\text{app}}(c) = D_0\,(1 + k_D\,c) \tag{19}$$

The intercept D_0 gives the infinite-dilution *R*h,0; the slope gives the diffusion interaction parameter *k*D (positive in good solvents, negative in poor ones).

> Both are run from the **DLS tab** method dropdown ("Γ vs q^2 (sample, multi-angle)" and "D vs c (sample, multi-conc.)"): they operate on **all** of the selected sample's DLS measurements (of the current Mw fraction), not just the one in view.

10.4 Depolarized DLS (DDLS): rotational diffusion

For an **optically anisotropic** scatterer (a rigid rod, or an anisotropic particle), the **depolarized** (VH) correlogram decays faster than the polarized (VV) one because molecular *reorientation* adds to the decay. In the small-qL regime the two field (g_1) decay rates are

$$\Gamma_{\text{VV}} = q^2 D_t, \qquad \Gamma_{\text{VH}} = q^2 D_t + 6D_r, \tag{42}$$

where *D*t is the translational and *D*r the **rotational** diffusion coefficient [9], [10]. Subtracting isolates the rotation, and its inverse is the reorientation time:

$$D_r = \frac{\Gamma_{\text{VH}} - \Gamma_{\text{VV}}}{6}, \qquad \tau_{\text{rot}} = \frac{1}{6D_r}. \tag{43}$$

Each correlogram is fitted for its Γ with the cumulant engine (§10.1); since a homodyne *intensity* correlogram decays at **2**Γ, the field rate Γ is what enters Eqs. (42)–(43) — the factor of 2 is handled by the fitter, not double-counted here. Across angles, *D*t is taken from the **VV** channel (ΓVV vs q^2 through the origin — the whole VV signal, cleanly measured), and *D*r from the mean of the per-angle differences. The 6*D*r term is q-independent, so a plot of ΓVH vs q^2 is a line of slope *D*t and intercept 6*D*r — the translational/rotational separation at a glance.

Validity. Eq. (42) is a single exponential only for **qL** $\lesssim$ **3** (above it, higher rotational modes and intramolecular interference appear; [10]). If the rod length L is supplied, qL is evaluated per angle and the result is flagged when it exceeds the limit.

Uncertainty. A single angle (one VV + one VH correlogram) carries **no** ± (one correlogram per channel; [11]). Across angles the points are independent, so *D*t gets the through-origin regression SE and *D*r the spread (SD/$\sqrt{n}$) of the per-angle estimates — both labeled statistical-only (§15).

Practical note. The VH signal is intrinsically **weak** (the matrix/solvent scatters polarized), so depolarized acquisition needs a clean analyzer and long counting; flexible polymer coils barely depolarize, so DDLS targets rods and anisotropic particles. Mapping *D*t, *D*r to a rod's length and diameter (Tirado–García de la Torre) is described in §10.5.

> Run from the **DLS tab** → **Depolarized (DDLS)** sub-tab. Tag each correlogram's polarization (VV/VH) in the **Data tab** (the *Polarization* row); the sub-tab pairs a VV with a VH at each angle automatically. **Replicates** — several VV (or VH) correlograms at the *same* angle — are averaged (their fitted Γ values are meaned) rather than one being dropped; for weak VH, pre-averaging the correlograms themselves via the sidebar *Average correlation functions* action (then tagging the result) gives better signal-to-noise.

10.5 DDLS shape models: dimensions from Dt and Dr

Dt and Dr are **shape-free observables**; turning them into particle *dimensions* requires assuming a geometry, so the outputs below are **model-dependent** — dimensions of an *assumed* shape, never a direct measurement. Two models are offered, and read together they diagnose the shape.

Sphere (one unknown). The rotational diffusion of a sphere of radius R is the Stokes–Einstein–Debye relation, whose inverse gives R:

$$D_r = \frac{k_B T}{8\pi\eta R^3}, \qquad R = \left(\frac{k_B T}{8\pi\eta D_r}\right)^{1/3}. \tag{44}$$

For a *true* sphere this equals the translational (Stokes) radius $R\text{h} = k\text{B}\,T/(6\pi\eta D\text{t})$, so the program reports both and their ratio $R(D\text{r})/R\text{h}$: ≈ 1 means a sphere is plausible; a ratio above 1 is the **rod signature** (rotation reports a smaller effective size than translation), flagging that the sphere model does not apply.

Rigid rod (two unknowns). For a cylinder of length L and diameter d (axial ratio $p = L/d$), the Tirado–García de la Torre [12] relations are

$$D_t = \frac{k_B T}{3\pi\eta L}(\ln p + \nu), \qquad D_r = \frac{3k_B T}{\pi\eta L^3}(\ln p + \delta_\perp), \tag{45}$$

$$\nu = 0.312 + \tfrac{0.565}{p} - \tfrac{0.100}{p^2}, \qquad \delta_\perp = -0.662 + \tfrac{0.917}{p} - \tfrac{0.050}{p^2}. \tag{46}$$

With two observables and two unknowns the rod is exactly determined. There is no closed-form inverse, so the program solves it numerically: at a trial p, Dt fixes L (Eq. 45, left); the predicted Dr is monotonic in p, so a one-dimensional root find recovers p, then L and $d = L/p$.

Validity. The end-effect polynomials (Eq. 46) are fitted only for $\mathbf{2 < p < 30}$; a recovered p outside that range is flagged as an extrapolation, and if no rod can reproduce both coefficients the rod model is reported as not fitting.

Uncertainty. The sphere radius SE follows from Dr by the delta method ($R \propto D\text{r}{-1/3} \Rightarrow \text{SE}(R)/R = 1/3{\cdot}\text{SE}(D\text{r})/D\text{r}$). The rod inverse is nonlinear, so its L, d SEs are propagated by **Monte Carlo** — sampling Dt, Dr from their SEs and inverting each — which is the sampling spread directly, with no linearization to validate (§15).

Mapping diffusion to dimensions is the heart of why DDLS is run on rods and anisotropic particles. References: Tirado, López Martínez and García de la Torre [12] (rod); Balog et al. [13] (depolarized DLS of anisotropic nanoparticles, the sphere context).

> Run from the same **DLS tab → Depolarized (DDLS)** sub-tab: both models compute automatically after *Run DDLS* (the sphere needs the solvent viscosity).

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11. Static Light Scattering (SLS) Analysis

The SLS module extracts the weight-average molecular weight Mw, the radius of gyration Rg, and the second virial coefficient A_2. It first converts raw intensities to absolute excess Rayleigh ratios, then constructs the Zimm, Berry, or Debye plots.

11.1 Calibration to an Excess Rayleigh Ratio

Raw intensities are placed on an absolute scale in steps: subtract the detector dark count, subtract the pure-solvent scattering (to isolate the excess from the polymer), correct for the scattering volume via $\sin\theta$, apply the refractive-index correction Eq. (5), and scale by a calibration constant kc:

$$\Delta R_\theta = k_c \, \sin\theta \, (I_{\text{sample}} - I_{\text{solvent}}) \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}} \right)^2 \tag{20}$$

There is **one** calibration model, used identically for any instrument. The program computes its own kc from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's Rayleigh factor (§7.5, in the matching geometry):

$$k_c = \frac{R_{\text{std}}}{I_{\text{cal,net}} \, \sin\theta_{\text{cal}}} \tag{21}$$

The constant is shown and can be edited. Any value stored in the instrument file is kept only for reference and **never** enters the calculation (a stored kc goes stale after a re-alignment, and vendor Rayleigh factors are often outdated).

> **Uncalibrated fallback.** If no kc can be computed or supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated. Then Mw and the absolute A_2 are marked unreliable — but Rg is unaffected (it comes from the ratio of slope to intercept, where the scale cancels), and so is the calibration-free product $2A_2M$w (§11.4). Those remain valid without calibration.

11.2 Single-Concentration (Apparent) Analysis

At one concentration, a **Debye** plot of $Kc/\Delta R\theta$ against q^2 is linear; the intercept gives an apparent molecular weight and the slope an apparent radius of gyration. These are **apparent** because the intercept still contains the concentration term,

$$\left. \frac{Kc}{\Delta R_\theta} \right|_{q\to 0} = \frac{1}{M_w} + 2A_2 c, \tag{22}$$

i.e. $1/M$w + $2A_2c$, not $1/M$w. A single angle yields only an apparent Mw and no Rg. Apparent values are never presented as final.

11.3 Zimm and Berry Double Extrapolation

The full analysis combines several concentrations and angles and extrapolates to both zero angle and zero concentration. The first-order Zimm equation,

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{M_w}\left(1 + \frac{q^2 R_g^2}{3}\right) + 2A_2 c, \tag{23}$$

is linear in (q^2, c), so a single global regression $Kc/\Delta R\theta = a + b\ q^2 + d\ c$ gives all three at once:

$$M_w = \frac{1}{a}, \qquad R_g = \sqrt{\frac{3b}{a}}, \qquad A_2 = \frac{d}{2}. \tag{24}$$

The **Berry** construction uses the square-root ordinate $\sqrt{Kc/\Delta R_\theta} = a\text{'} + b\text{'}q^2 + d\text{'}c$, giving

$$M_w = \frac{1}{a'^2}, \qquad R_g = \sqrt{\frac{6b'}{a'}}, \qquad A_2 = d'\, a'. \tag{25}$$

Berry linearizes the high-qRg regime (roughly Mw above ~1 MDa, or where qRg approaches 1) better than Zimm. The program reports which construction it used, and the result is marked **thermodynamic**, not apparent.

> **Spacing constant.** The classic grid spreads each concentration's points along the x-axis by $q^2 + k\ c$, where k is a purely **aesthetic** spacing constant. Set it in the **Spacing k** box (or press **Suggest** for the automatic value, q^2max/$c$max); the chosen k appears in the x-axis label. It only changes how spread-out the curves look — Mw, Rg, and A_2 come from the real (q^2, c) fit and are unaffected.

11.4 Calibration-Free Second Virial Coefficient

A_2 can be obtained without any absolute calibration, from ratios of excess intensities across concentration at a fixed angle. Because only ratios enter, the optical constant K and the Rayleigh-factor calibration cancel:

$$Y(c) = \frac{I(c_{\text{ref}})/c_{\text{ref}}}{I(c)/c}, \qquad \frac{\text{slope}}{\text{intercept}} = 2A_2 M_w. \tag{26}$$

This yields the product $2A_2M$w directly; if Mw is known, A_2 follows. It is valuable when absolute calibration is uncertain, as in low-dn/dc systems.

11.5 Guinier Plot

A single-concentration route to Rg using the **log intensity** — the counterpart to the Debye plot of §11.2. The Guinier approximation ΔR(q) = ΔR(0)·exp(−q^2 Rg^2/3) linearizes as

$$\ln[\Delta R(q)] = \ln[\Delta R(0)] - \frac{R_g^2}{3}\, q^2, \tag{27}$$

so a straight-line fit of $\ln(\Delta R)$ vs q^2 gives $R\mathrm{g} = \sqrt{-3\,\mathrm{slope}}$ and an apparent $M\mathrm{w}$ from the intercept ($\Delta R(0) = Kc\, M\mathrm{w}$). Like Debye's, the Guinier $R\mathrm{g}$ is **scale-independent** (it comes from the slope of a log-intensity plot), so it is reliable even on an **uncalibrated** run; only the intercept-derived $M\mathrm{w}$ needs calibration. Valid in the Guinier regime $qR\mathrm{g} \lesssim 1.3$ (the program flags the largest $qR\mathrm{g}$ used); for larger species, Berry or Zimm linearize the high-$qR\mathrm{g}$ regime better. Reference: [14] (origin); see [1] and [15] for accessible derivations of the Guinier plot.

11.6 Depolarization Correction (Anisotropic Scatterers)

An **optically anisotropic** scatterer — one whose polarizability differs along different molecular axes (rigid rods, many nanoparticles, small anisotropic molecules like toluene; flexible coils are very nearly isotropic) — scrambles the polarization of part of the light it scatters. With the **vertically polarized** incident light used by essentially every modern instrument, this appears as a horizontally polarized (*depolarized*) component. Measuring it both enables the **Cabannes correction** that keeps $M\mathrm{w}$ from being overestimated, and yields the **optical anisotropy** as a result in its own right. This is the **static** side of depolarized light scattering; the **dynamic** side (rotational diffusion $D\mathrm{r}$ from the depolarized correlogram) is described in §10.4.

Convention. Two-letter subscripts are **(incident, analyzer)**: $I\mathrm{VV}$ = vertical incident + vertical analyzer (the ordinary polarized intensity); $I\mathrm{VH}$ = vertical incident + horizontal analyzer (the depolarized intensity). The **depolarization ratio** and its natural-light equivalent are

$$\rho_v = \frac{I_{\mathrm{VH}}}{I_{\mathrm{VV}}}, \qquad \rho_u = \frac{2\rho_v}{1+\rho_v}, \tag{38}$$

where $\rho\mathrm{u}$ is the ratio for **unpolarized** incident light used by the classical literature (the conversion is exact). Physically $0 \le \rho\mathrm{v} \le 3/4$: $\rho\mathrm{v} = 0$ is an isotropic scatterer, $\rho\mathrm{v} = 3/4$ the fully anisotropic limit. A value outside this range signals **stray depolarized light** (polarizer/analyzer leakage, reflections, or an over-subtracted dark count) and is flagged, not used.

Cabannes correction (vertical incidence). The anisotropic part inflates the measured VV Rayleigh ratio, so an uncorrected $M\mathrm{w}$ comes out too high. The **isotropic** excess Rayleigh ratio — the one the Zimm/Debye/Berry analysis (§11.2–11.3) should use — is recovered by

$$\Delta R_{\mathrm{iso}} = \Delta R_{\mathrm{VV}} \left(1 - \tfrac{4}{3}\,\rho_v\right). \tag{39}$$

For an isotropic scatterer ($\rho\mathrm{v} = 0$) this is the identity, so applying it is always safe. The removed anisotropic fraction is $(4/3)\rho\mathrm{v}$.

Optical anisotropy. In Chu's parametrization the apparent molecular weight and the depolarized Rayleigh ratio carry the mean-square optical anisotropy δ^2,

$$M_{\mathrm{app}} = M_w\big(1 + \tfrac{4}{5}\delta^2\big), \qquad \delta^2 = \frac{5\rho_v}{3-4\rho_v}, \tag{40}$$

so δ^2 is 0 for an isotropic scatterer and grows without bound as $\rho\mathrm{v} \to 3/4$. Eliminating δ^2 between $M\mathrm{app}$ and $\rho\mathrm{v}$ reproduces Eq. (39) exactly — the two routes agree. (δ^2 here is Chu's normalization; convert between anisotropy conventions through the depolarization ratio, which is convention-independent, not by equating differently-normalized anisotropies.)

Natural-light form (for older datasets). For **unpolarized** incident light the classical Cabannes factor recovers the isotropic part from the total at 90°,

$$\Delta R_{\text{iso}} = \Delta R_{\text{total}} \cdot \frac{6 - 7\rho_u}{6 + 6\rho_u}. \tag{41}$$

This is provided for reading natural-light data; modern instruments use Eq. (39).

Uncertainty. A single VV/VH intensity pair carries **no** statistical SE on ρv (consistent with the project rule that one measurement is not an ensemble; §15). When **replicate** intensities are available, the SE on ρv is propagated as a ratio of independent quantities (Eq. 34); δ^2 and the Cabannes factor then follow by the delta method (Eq. 33).

References: Chu [1], §8.4.1.A, Eqs. 8.4.7–8.4.10 (vertical-incidence correction, δ^2); Coumou, Mackor and Hijmans [16] (natural-light Cabannes factor); Sivokhin and Kazantsev [3] (toluene ρv; ρu ↔ ρv).

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15. Uncertainty Estimates

An uncertainty is part of an analysis, not decoration. The program reports the **statistical** (regression) uncertainty of a fit — and **only** where the fitted points are genuinely independent measurements. Where no honest uncertainty exists, **none is reported**. Two scope rules follow from this:

1. **Statistical only.** Every ± below is the spread implied by the scatter of the fitted points. It **excludes systematic error** — calibration (the constant kc) and dn/dc in particular. Because Mw $\propto 1/(\mathrm{d}n/\mathrm{d}c)^2$ and scales with kc, those systematics usually dominate the true error of an absolute Mw or A_2; the displayed ± is a **lower bound** on the total uncertainty and is labelled *"± statistical; excludes calibration & dn/dc"* wherever a calibration-dependent quantity is shown. Rg, Rh, the size exponent ν, and the calibration-free $2A_2M$w are scale-independent, so their statistical ± is the whole statistical story.
2. **No uncertainty from a single correlogram.** A correlogram's lag channels are strongly correlated [7], [11], so an ordinary least-squares covariance from **one** correlogram assumes independence it does not have and **under-reports** the truth. ISO 22412 [17] therefore derives DLS dynamic uncertainty from **repeat measurements**. Accordingly the single-correlogram dynamic fits (cumulant, single/double exponential, KWW, lognormal) report **no** ±. The honest dynamic ± instead comes from **replicate averaging** (§15.4): when you average true repeats, the spread of the per-replicate fits gives an ISO 22412 SD/SEM. The cumulant PDI and the fit RMS remain as width/goodness diagnostics.

15.1 The estimator: HC3 heteroscedasticity-consistent standard errors

For a linear model $\mathbf{y} = \mathbf{X}\mathbf{b}$ with n points, p parameters and residuals ei, the parameter covariance is the **HC3 "sandwich"** estimator [18], [19]:

$$\mathrm{Cov}(\mathbf{b}) = (\mathbf{X}^\top\mathbf{X})^{-1}\left[\sum_i \frac{e_i^2}{(1-h_{ii})^2}\,\mathbf{x}_i\mathbf{x}_i^\top\right](\mathbf{X}^\top\mathbf{X})^{-1}, \qquad h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i. \tag{30}$$

Unlike the textbook $s^2(\mathbf{X}^\top\mathbf{X})^{-1}$, HC3 does **not** assume the points share one error variance, so it does not under-report when point precision varies across angles or concentrations (e.g. a per-angle Γ whose precision changes with q). It reduces to the ordinary formula under equal-variance errors and is mildly **conservative** at small n (Monte-Carlo-verified here to meet or exceed the true sampling spread — it never under-reports). When the residual degrees of freedom are too few ($n - p < 2$) no SE is reported, since one residual cannot honestly support a spread (and HC3 blows up as $n - p \to 1$). The standard error of a coefficient is the square root of its diagonal, e.g. for a straight line $y = a + b\ x$:

$$\mathrm{SE}(b) = \sqrt{\mathrm{Cov}_{bb}}, \qquad \mathrm{SE}(a) = \sqrt{\mathrm{Cov}_{aa}}. \tag{31}$$

A through-origin slope $y = b\ x$ (used for $\Gamma = D\ q^2$) uses the same sandwich with the one-column design $\mathbf{X} = \mathbf{x}$, giving SE(b) from the scalar covariance.

The Zimm/Berry global fit is multilinear — ordinate $= a + b\ q^2 + d\ c$ with design $\mathbf{X} = [1,\ q^2,\ c]$ — and uses the same Cov($\mathbf{b}$) of Eq. (30) for the full 3×3 covariance:

$$\text{ordinate} = a + b\,q^2 + d\,c, \qquad \mathbf{X} = [\,\mathbf{1}\ \ \mathbf{q^2}\ \ \mathbf{c}\,]. \tag{32}$$

15.2 Propagating a covariance to a derived quantity (delta method)

A scalar function $f(\mathbf{b})$ of the fitted coefficients has, to first order, the standard error

$$\mathrm{SE}(f) = \sqrt{\mathbf{J}^\top\,\mathrm{Cov}(\mathbf{b})\,\mathbf{J}}, \qquad \mathbf{J} = \nabla_{\mathbf{b}} f. \tag{33}$$

This delivers every SLS SE from Eq. (30)/(32) with the appropriate Jacobian $\mathbf{J}$:

Quantity	Relation	Jacobian $\mathbf{J}$ (order of params)
Mw (Zimm)	$M\mathrm{w} = 1/a$	$[-1/a^2,\ 0,\ 0]$
A_2 (Zimm)	$A_2 = d/2$	$[0,\ 0,\ 1/2]$
Rg (Zimm)	$R\mathrm{g} = \sqrt{3b/a}$	$[-R\mathrm{g}/(2a),\ R\mathrm{g}/(2b),\ 0]$
Mw (Berry)	$M\mathrm{w} = 1/a'^2$	$[-2/a'^3,\ 0,\ 0]$
A_2 (Berry)	$A_2 = d'a'$	$[d',\ 0,\ a']$
Mwapp (Debye)	= 1/intercept	[−1/intercept^2, 0]
Rgapp (Debye)	$= \sqrt{3\,\text{slope}/\text{intercept}\cdot M_w}$	[−Rg/(2·intercept), Rg/(2·slope)]
Rg (Guinier)	from slope of ln ΔR vs q^2	SE(Rg) = (3/2Rg)·SE(slope)
$2A_2M$w (cal-free)	= slope/intercept	[−slope/intercept^2, 1/intercept]

For a **ratio** of two *independent* quantities $r = a/b$ (used for $\rho = R\mathrm{g}/R\mathrm{h}$, with Rg from SLS and Rh from DLS — measured independently), Eq. (33) reduces to the familiar fractional sum in quadrature:

$$\mathrm{SE}(r) = \left|\frac{a}{b}\right| \sqrt{\left(\frac{\mathrm{SE}(a)}{a}\right)^2 + \left(\frac{\mathrm{SE}(b)}{b}\right)^2}. \tag{34}$$

ρ is given a $\pm$ **only when both** Rg and Rh carry an SE (i.e. both came from a regression — a Zimm/Berry Rg with a Γ-q^2 or D-c Rh); a single-cumulant Rh has no SE, so ρ then prints without $\pm$.

For a **power map** $f = k\ x$p (used for Rh $\propto D^{-1}$, so $p = -1$), Eq. (33) gives a fractional relation:

$$\frac{\mathrm{SE}(f)}{|f|} = |p|\,\frac{\mathrm{SE}(x)}{|x|}, \qquad \text{so } \frac{\mathrm{SE}(R_h)}{R_h} = \frac{\mathrm{SE}(D)}{D}. \tag{35}$$

The cross-sample scaling exponents (ν in Rg/Rh $\propto M$wν, the slope in $A_2 \propto M$wslope) are slopes of a log–log regression, so their SE is Eq. (31) applied to (log Mw, log y).

15.3 A consistency check, not an uncertainty: the Zimm two-route *M*w

The Zimm grid is extrapolated to the intercept along two independent routes — the $c \to 0$ line and the $q \to 0$ line — which must meet at the same $1/M$w. The program reports both intercept-derived weights and their relative gap:

$$\text{agreement} = \frac{\left|M_w^{(c\to 0)} - M_w^{(q\to 0)}\right|}{\frac{1}{2}\left(M_w^{(c\to 0)} + M_w^{(q\to 0)}\right)}. \tag{36}$$

This is a **diagnostic of extrapolation quality** (curvature, an over-masked grid), not a statistical uncertainty; a gap above ~10 % is flagged for inspection (the interface phrases it as the two routes "differ by X %"). On the Zimm/Berry plot both extrapolation lines are drawn **all the way to the axis** ($q^2 \to 0$ and $c \to 0$), meeting at the common intercept ($\approx 1/M$w, marked), so the double extrapolation is visible end to end.

15.4 Repeat measurements: the one DLS dynamic uncertainty (ISO 22412)

Because a single correlogram cannot supply an honest dynamic $\pm$ (rule 2), the program gets one the way ISO 22412 [17] prescribes — from **repeat measurements**. Select a set of true replicates (the same sample re-measured back-to-back, e.g. a 10-run repeat set) in the navigator, right-click, and choose **Average derived results**. Each replicate is fitted independently with the method you pick, and for a parameter x (e.g. Rh) the n fitted values are summarised as a mean and a standard error of the mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \mathrm{SEM} = \frac{s}{\sqrt{n}}. \tag{37}$$

The reported value is $\bar{x} \pm$ SEM (none for $n < 2$, where no spread is defined). Unlike a within-correlogram covariance, the xi from separate runs **are** independent draws, so Eq. (37) does not under-report; it also captures genuine run-to-run drift (alignment, dust, number fluctuations) that one curve cannot see. The averaged Rh $\pm$ SEM is written into the sample (so $\rho = R$g/Rh can use it), never overwriting a hand-entered Rh. This SE was Monte-Carlo validated: across many synthetic n-replicate sets the reported SD/$\sqrt{n}$ matched the actual SD of the set-means to ≈ 1 % and the mean estimate was unbiased.

The companion action, **Average correlation functions**, instead takes the channel-by-channel mean of the replicate $g_2(\tau) - 1$ curves (requiring an identical lag grid) to produce a cleaner *point-estimate* correlogram you can fit like any measurement; it carries no ± of its own — the ± comes from Eq. (37).

Multi-peak (distribution) methods. *Average derived results* also accepts the distribution methods (NNLS / CONTIN / lognormal), which resolve a *variable* number of size-distribution peaks. These are averaged **by order of appearance**: each replicate's peaks are sorted by Rh, and position 1 (smallest Rh) is averaged across all runs, position 2 across all, etc., each reported as mean ± SD/$\sqrt{n}$ (Eq. 37) with its mean intensity weight and how many of the N runs resolved it. Two cautions are built in: (i) when runs **disagree on the peak count** a warning is shown, because a run that misses a peak shifts the positional alignment (inspect the distribution overlay before trusting the per-peak averages); and (ii) NNLS/CONTIN are **ill-posed** — their peak *positions* depend on the regularization and their fine structure is not always reproducible — so the cross-run ± is an honest **reproducibility spread**, not a calibrated error, and these peaks are **report-only** (never written to the sample's Rh). To get an averaged multi-population Rh that *can* feed ρ, average the correlograms (above) and fit that single clean curve with CONTIN in the usual way.

15.5 Where uncertainties are and are not reported

Reported (independent points → regression SE valid): SLS Zimm/Berry Mw, Rg, A_2; Debye/Guinier apparent Mw, Rg; calibration-free $2A_2M$w (and A_2 if Mw is supplied); Γ-vs-q^2 D and Rh; D-vs-c D_0, Rh,0, kD; the scaling exponents; $\rho = R\mathrm{g}/R\mathrm{h}$ when both inputs have an SE; and the **replicate-averaged DLS dynamic** parameters (Eq. 37 — the one single-angle dynamic ±).

Not reported (no honest statistical uncertainty): every single-correlogram dynamic fit (rule 2 above), *unless* repeats are averaged via §15.4; the NNLS/CONTIN distributions (ill-posed inversions — no defensible covariance; the "fine structure is not always reproducible" caveat stands); and a single-angle Mw (one datum, no fit).

15.6 How the ± is written

The displayed value's precision is set by its uncertainty (PDG convention): the SE is rounded to one significant figure — two if its leading digit is 1 — and the value is rounded to the same decimal place, e.g. $1.234\times10^6 \pm 5.6\times10^4 \rightarrow$ **$(1.23 \pm 0.06)\times10^6$**, and 40.5 ± 0.83 nm → **40.5 ± 0.8 nm**.

> **Standing rule.** Every analysis — existing or newly added, including the planned depolarized scattering (DPLS) module — must state how it treats uncertainty and, when it reports one, document the formula here with a numbered equation, exactly as the analysis itself is documented.

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References

[1] B. Chu, *Laser light scattering: Basic principles and practice*, 2nd ed. Academic Press, 1991.

[2] K. Takahashi, A. Takano, S. Kinugasa, and H. Sakurai, "Determination of the rayleigh ratio of toluene with an uncertainty analysis," *Analytical Sciences*, vol. 35, no. 9, pp. 1045–1051, 2019, doi: 10.2116/analsci.19P103.

[3] A. A. Sivokhin and O. A. Kazantsev, "Temperature dependence of the rayleigh ratio of toluene and its depolarization ratio," *ChemistrySelect*, vol. 6, no. 35, pp. 9499–9502, 2021, doi: 10.1002/slct.202102196.

[4] M. Rubinstein and R. H. Colby, *Polymer physics*. Oxford University Press, 2003.

[5] L. J. Fetters, N. Hadjichristidis, J. S. Lindner, and J. W. Mays, "Molecular weight dependence of hydrodynamic and thermodynamic properties for well-defined linear polymers in solution," *Journal of Physical and Chemical Reference Data*, vol. 23, no. 4, pp. 619–640, 1994, doi: 10.1063/1.555949.

[6] D. E. Koppel, "Analysis of macromolecular polydispersity in intensity correlation spectroscopy: The method of cumulants," *Journal of Chemical Physics*, vol. 57, no. 11, pp. 4814–4820, 1972, doi: 10.1063/1.1678153.

[7] B. J. Frisken, "Revisiting the method of cumulants for the analysis of dynamic light-scattering data," *Applied Optics*, vol. 40, no. 24, pp. 4087–4091, 2001, doi: 10.1364/AO.40.004087.

[8] S. W. Provencher, "CONTIN: A general purpose constrained regularization program for inverting noisy linear algebraic and integral equations," *Computer Physics Communications*, vol. 27, no. 3, pp. 229–242, 1982, doi: 10.1016/0010-4655(82)90174-6.

[9] R. Pecora, "Doppler shifts in light scattering from pure liquids and polymer solutions," *Journal of Chemical Physics*, vol. 40, no. 6, pp. 1604–1614, 1964, doi: 10.1063/1.1725368.

[10] K. Zero and R. Pecora, “Rotational and translational diffusion in semidilute solutions of rigid-rod macromolecules,” *Macromolecules*, vol. 15, no. 1, pp. 87–93, 1982, doi: 10.1021/ma00229a018.

[11] K. Schätzel, “Noise on photon correlation data: I. Autocorrelation functions,” *Quantum Optics*, vol. 2, no. 4, pp. 287–305, 1990, doi: 10.1088/0954-8998/2/4/002.

[12] M. M. Tirado, C. López Martínez, and J. García de la Torre, “Comparison of theories for the translational and rotational diffusion coefficients of rod-like macromolecules. Application to short DNA fragments,” *Journal of Chemical Physics*, vol. 81, no. 4, pp. 2047–2052, 1984, doi: 10.1063/1.447827.

[13] S. Balog *et al.*, “Characterizing nanoparticles in complex biological media and physiological fluids with depolarized dynamic light scattering,” *Nanoscale*, vol. 7, no. 14, pp. 5991–5997, 2015, doi: 10.1039/c4nr06538g.

[14] A. Guinier, “La diffraction des rayons x aux très petits angles: Application à l’étude de phénomènes ultramicroscopiques,” *Annales de Physique (Paris)*, vol. 11, no. 12, pp. 161–237, 1939, doi: 10.1051/anphys/193911120161.

[15] P. S. Russo, K. A. Streletzky, W. Huberty, X. Zhang, and N. Edwin, “Characterization of polymers by static light scattering,” in *Molecular characterization of polymers: A fundamental guide*, M. I. Malik, J. Mays, and M. R. Shah, Eds., Ch. 13 in Molecular Characterization of Polymers: A Fundamental Guide, eds. Malik, Mays & Shah (Elsevier); Elsevier, 2021, ch. 13.

[16] D. J. Coumou, E. L. Mackor, and J. Hijmans, “Isotropic light scattering in pure liquids,” *Transactions of the Faraday Society*, vol. 60, pp. 1539–1547, 1964, doi: 10.1039/TF9646001539.

[17] International Organization for Standardization, “Particle size analysis – dynamic light scattering (DLS).” ISO 22412:2017, Geneva, 2017.

[18] J. G. MacKinnon and H. White, “Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties,” *Journal of Econometrics*, vol. 29, no. 3, pp. 305–325, 1985, doi: 10.1016/0304-4076(85)90158-7.

[19] J. S. Long and L. H. Ervin, “Using heteroscedasticity consistent standard errors in the linear regression model,” *The American Statistician*, vol. 54, no. 3, pp. 217–224, 2000, doi: 10.1080/00031305.2000.10474549.